

4(a)	(two) waves travelling (at same speed) in opposite directions overlap	B1
	waves (are same type and) have same frequency/wavelength	B1
4(b)(i)	$\lambda = 12 / 250 (= 0.048 \text{ m})$	C1
	distance = 1.5×0.048 = 0.072 m	A1
4(b)(ii)	$T = 1 / 250$ = 0.004 (s) or 4 (ms)	C1
	1. curve drawn is mirror image of that in Fig. 4.2 and labelled P	A1
	2. horizontal line drawn between A and B and labelled Q	A1

Question	Answer	Marks
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